

Medvale Internal Medicine

Chapter 3I

Gastroenterology and Hepatology Review: Integrated Cases, Diagnostic Algorithms, and High-Yield Tables

Original study resource. Source materials and prior chapters were used as coverage signals only; wording, tables, and diagrams are newly authored for Medvale.

This closing GI chapter is designed as a synthesis layer for graph-RAG extraction. It links symptoms to mechanisms, disease scripts, diagnostic tests, treatments, complications, and common pitfalls across the esophagus, stomach, small bowel, colon, liver, biliary tree, pancreas, and appendix. It intentionally repeats important numeric anchors because retrieval systems perform better when clinically meaningful associations appear in prose, tables, and case explanations.

1. How to Use This Review Chapter

The previous GI chapters built the organ-based content: esophagus and stomach, small bowel, colon and rectum, inflammatory and functional bowel disease, liver and portal hypertension, hepatobiliary-pancreatic disease, and viral/metabolic/toxic liver injury. This chapter reorganizes that content by the way patients present. Internal medicine decisions usually begin with a problem such as abdominal pain, hematemesis, chronic diarrhea, jaundice, ascites, or an abnormal imaging result. A strong GI illness script therefore connects four layers: symptom pattern, pathophysiology, localization, and first management move.

For exam reasoning, begin by deciding whether the patient is unstable. Hemodynamic instability, peritonitis, altered mental status, severe sepsis, acute liver failure, toxic megacolon, bowel ischemia, cholangitis, and perforation override ordinary outpatient algorithms. In unstable patients, resuscitation and source control usually occur before complete etiologic certainty. In stable patients, the goal is to use the symptom pattern to choose the test that most directly changes management.

For graph RAG, this chapter should be treated as a hub. Each disease is not only a node with a definition; it is also connected to triggers, risk factors, exam clues, laboratory anchors, imaging choices, treatment decisions, complications, and mimic diagnoses. For example, melena links to upper GI bleeding, iron deficiency anemia, peptic ulcer disease, varices, small bowel bleeding, colon cancer in selected cases, BUN elevation, endoscopy, transfusion threshold, and PPI therapy. The same concept should be discoverable from several different clinical directions.

2. GI Red Flags: When the Algorithm Starts with Stabilization

Many GI disorders are benign or chronic, but the dangerous presentations are time-sensitive because delayed management can produce shock, perforation, sepsis, liver failure, or irreversible bowel infarction. A patient with severe abdominal pain out of proportion to the exam may have acute mesenteric ischemia even when the abdomen is initially soft. A patient with RUQ pain, fever, jaundice, hypotension, and confusion has ascending cholangitis until proven otherwise. A cirrhotic patient with ascites and fever, abdominal pain, renal injury, or altered mental status needs diagnostic paracentesis because spontaneous bacterial peritonitis can be subtle and lethal.

Red flag	Major diagnoses to consider	Immediate move
Hypotension, tachycardia, syncope, ongoing hematemesis or hematochezia	Upper or lower GI bleeding, variceal hemorrhage, ischemic colitis	Airway assessment, two large-bore IVs, fluids/blood, type and cross, urgent endoscopy or localization
Rigid abdomen, rebound, guarding, free air	Perforated ulcer, perforated diverticulitis, ischemic bowel, ruptured viscus	NPO, IV fluids, broad antibiotics, urgent surgery consult
Severe pain out of proportion to exam	Acute mesenteric ischemia	Check lactate, CT angiography when stable, broad antibiotics, anticoagulation if appropriate, urgent vascular/surgery
Jaundice plus fever or sepsis	Ascending cholangitis	Blood cultures, IV antibiotics, fluids, biliary decompression by ERCP/PTC after stabilization
Encephalopathy plus INR >1.5 without known cirrhosis	Acute liver failure	ICU/transplant center, identify cause, protect airway if needed, treat cerebral edema risk
Toxic megacolon signs	Severe UC, C. difficile, ischemic or infectious colitis	Stop antimotility drugs, fluids, antibiotics/steroids by cause, surgical monitoring

3. Integrated Diagnostic Algorithms

Algorithm A: Acute Abdominal Pain in Adults

Step	Decision	Action
1	First ask: unstable or peritoneal?	If hypotension, sepsis, rigid abdomen, rebound, guarding, or free air is present, resuscitate and obtain urgent surgical evaluation.

Step	Decision	Action
2	Localize pain	RUQ suggests biliary disease, hepatitis, hepatic abscess, or pneumonia; epigastric pain suggests PUD, pancreatitis, MI, biliary colic; RLQ suggests appendicitis, Crohn ileitis, ovarian/testicular mimics; LLQ suggests diverticulitis, ischemic colitis, gynecologic or urinary disease.
3	Pattern and timing	Colicky pain suggests obstruction or biliary colic. Constant progressive pain suggests inflammation, ischemia, perforation, pancreatitis, or abscess.
4	Choose first imaging	RUQ ultrasound for suspected gallbladder disease; CT abdomen/pelvis with contrast for appendicitis, diverticulitis, obstruction, abscess, malignancy, or unclear severe pain; CT angiography for suspected mesenteric ischemia.
5	Do not anchor prematurely	GI pain can be cardiac, pulmonary, vascular, urinary, gynecologic, metabolic, or medication-related. ECG, pregnancy test, urinalysis, and chest imaging may be appropriate depending on the story.

Algorithm B: Upper GI Bleeding

Step	Decision	Action
1	Recognize presentation	Hematemesis, coffee-ground emesis, melena, or brisk hematochezia with instability can all be upper GI bleeding.
2	Stabilize first	Assess airway, place two large-bore IVs, type/cross, CBC, CMP, PT/INR, platelets, and resuscitate. Transfuse most stable hospitalized patients at hemoglobin about 7 g/dL; use a higher threshold when shock or significant cardiovascular disease is present.
3	Start empiric therapy by likely source	PPI for suspected peptic ulcer bleeding; octreotide plus antibiotics for suspected variceal bleeding in cirrhosis; reverse significant coagulopathy when clinically needed.
4	Definitive test	Upper endoscopy is diagnostic and therapeutic, typically within 24 hours after stabilization; urgent endoscopy is needed for ongoing severe bleeding.
5	If endoscopy fails	Repeat endoscopy for recurrent ulcer bleeding when appropriate, interventional radiology embolization, or surgery for persistent instability or massive transfusion requirement.

Algorithm C: Lower GI Bleeding and Occult Blood

Step	Decision	Action
1	Differentiate hematochezia from melena	Hematochezia usually localizes to colon/rectum, but 5-10 percent of hematochezia can be brisk upper GI bleeding, especially with instability.
2	Initial evaluation	Vitals, rectal exam, medication review, CBC, coagulation tests, renal function, and type/cross if significant bleeding.
3	Stable patient	Colonoscopy after adequate preparation is usually the key diagnostic and therapeutic test for colonic sources.
4	Unstable active bleeding	Consider CT angiography or catheter angiography when rapid localization and embolization may be needed.
5	Occult blood or iron deficiency	In adults, especially age 45 and older or with anemia, evaluate for colorectal cancer and other bleeding lesions. If colonoscopy is unrevealing and suspicion remains, consider upper endoscopy and small bowel evaluation.

Algorithm D: Chronic Diarrhea

Step	Decision	Action
1	Classify stool	Watery diarrhea may be osmotic, secretory, functional, endocrine, bile acid related, or medication related. Fatty diarrhea suggests malabsorption or maldigestion. Bloody/inflammatory diarrhea suggests IBD, infection, ischemia, malignancy, or radiation injury.
2	Use red flags	Weight loss, anemia, nocturnal diarrhea, blood, fever, hypoalbuminemia, age of onset later in life, family history of IBD/CRC, and abnormal inflammatory markers argue against simple IBS.
3	Medication and diet review	Metformin, magnesium, PPIs, NSAIDs, antibiotics, laxatives, sugar alcohols, alcohol, and tube feeds can be causal.

Step	Decision	Action
4	Targeted tests	CBC, CMP, CRP, celiac serology, TSH when appropriate, stool pathogen testing when infectious risk exists, C. difficile testing after antibiotics/healthcare exposure, fecal calprotectin/lactoferrin for inflammation, fecal fat or elastase for malabsorption.
5	Endoscopy	Colonoscopy with biopsies is needed for suspected IBD, microscopic colitis, malignancy, unexplained inflammatory diarrhea, or persistent unexplained symptoms.

Algorithm E: Jaundice and Abnormal Liver Tests

Step	Decision	Action
1	Separate hepatocellular from cholestatic	AST/ALT predominant elevation suggests hepatocyte injury; alkaline phosphatase and direct bilirubin predominance suggests cholestasis or obstruction.
2	Assess severity	INR, bilirubin, albumin, mental status, platelets, renal function, and encephalopathy determine urgency more than AST/ALT height alone.
3	Image cholestasis	RUQ ultrasound is usually first to look for ductal dilation, gallstones, masses, or cirrhosis features. MRCP/ERCP is considered when obstruction or cholangitis is suspected.
4	Serologic workup	Viral hepatitis studies, autoimmune markers, iron studies, ceruloplasmin in younger patients, alpha-1 antitrypsin testing when appropriate, medication/toxin review, alcohol and metabolic risk assessment.
5	Emergency patterns	Acute liver failure is encephalopathy plus INR >1.5 without preexisting cirrhosis. Cholangitis is infection plus biliary obstruction. Both need rapid escalation.

Algorithm F: RUQ Pain and Biliary Syndromes

Step	Decision	Action
1	Biliary colic	Episodic postprandial RUQ/epigastric pain from transient cystic duct obstruction; no persistent fever or leukocytosis; ultrasound may show stones.
2	Acute cholecystitis	Persistent RUQ pain, fever, leukocytosis, Murphy sign; ultrasound shows stones, wall thickening, pericholecystic fluid, or sonographic Murphy sign; HIDA if ultrasound is equivocal.
3	Choledocholithiasis	CBD stone with jaundice or cholestatic labs; ultrasound may show ductal dilation; MRCP/EUS/ERCP selected by probability and need for therapy.
4	Cholangitis	Fever, jaundice, RUQ pain, and possible hypotension/altered mental status; treat sepsis and decompress the biliary tree.
5	Pancreatitis	Epigastric pain radiating to back plus lipase elevation; evaluate gallstones, alcohol, triglycerides, calcium, medications, and structural disease.

4. High-Yield Comparison Tables

Upper vs Lower GI Bleeding

Feature	Upper GI source	Lower GI source
Anatomic definition	Proximal to ligament of Treitz	Distal to ligament of Treitz; colon/rectum are common clinically
Classic presentation	Hematemesis, coffee-ground emesis, melena	Hematochezia, maroon stool, occult blood, iron deficiency
Can cause hematochezia?	Yes, if brisk and rapid	Yes, typical when distal/active
Common causes	PUD, erosive disease, varices, Mallory-Weiss tear, malignancy, Dieulafoy lesion	Diverticular bleeding, angiodysplasia, cancer/polyps, IBD, ischemic colitis, hemorrhoids
Key first procedure	EGD after stabilization	Colonoscopy when stable/prepped; CTA/angiography when unstable active bleeding
Medication clues	NSAIDs, aspirin, anticoagulants, antiplatelets, cirrhosis/portal HTN	Anticoagulants, antiplatelets, NSAIDs, radiation history, vascular disease

Crohn Disease vs Ulcerative Colitis

Feature	Crohn disease	Ulcerative colitis
Depth	Transmural inflammation	Mucosa/submucosa
Pattern	Skip lesions; patchy involvement	Continuous disease beginning in rectum
Location	Mouth to anus; terminal ileum common	Colon and rectum; backwash ileitis may occur
Symptoms	Nonbloody diarrhea, RLQ pain, weight loss, malabsorption, perianal disease	Bloody diarrhea, urgency, tenesmus, LLQ or diffuse cramps
Complications	Strictures, fistulas, abscesses, SBO, B12 deficiency, gallstones, kidney stones	Toxic megacolon, severe bleeding, colorectal cancer risk with duration/extent, PSC association
Smoking	Worsens risk/course	Classically protective association, but smoking is not recommended
Surgery	Not curative; recurrence common	Colectomy can cure colonic disease but extraintestinal issues may persist

IBS vs IBD vs Microscopic Colitis

Feature	IBS	IBD	Microscopic colitis
Mechanism	Gut-brain interaction, visceral hypersensitivity, motility and microbiome factors	Immune-mediated intestinal inflammation	Colonic mucosal inflammation visible on biopsy, often with normal gross colonoscopy
Bleeding	Absent	May occur, especially UC	Usually absent
Nocturnal symptoms	Uncommon	Can occur	Can occur
Weight loss/anemia	Absent unless another disorder	May occur	May occur but usually less dramatic
Testing	Clinical after red flags excluded	Fecal calprotectin, colonoscopy/biopsy, imaging	Colonoscopy with random biopsies
Typical treatment	Diet, reassurance, fiber/laxatives/antidiarrheals, antispasmodics, neuromodulators	5-ASA in selected UC, steroids for flares, biologics/immunomodulators, surgery when needed	Remove triggers; budesonide commonly used

Hepatocellular vs Cholestatic Injury

Pattern	Typical labs	Common diagnoses	First moves
Hepatocellular	AST/ALT disproportionately high compared with alkaline phosphatase	Viral hepatitis, alcohol-associated hepatitis, MASLD/MASH, ischemic hepatitis, DILI, autoimmune hepatitis, Wilson disease	Medication/toxin review, viral serologies, alcohol/metabolic assessment, autoimmune/inherited workup when indicated
Cholestatic	Alkaline phosphatase and direct bilirubin disproportionately high	Choledocholithiasis, cholangitis, PSC, PBC, pancreaticobiliary malignancy, infiltrative disease, drug cholestasis	Confirm hepatic source with GGT if needed; RUQ ultrasound; MRCP/ERCP when obstruction likely
Synthetic dysfunction	INR elevation, low albumin, encephalopathy, thrombocytopenia if portal HTN	Cirrhosis, acute liver failure, severe hepatitis, decompensated liver disease	Assess severity, manage complications, consider transplant referral if severe

Gallbladder, Bile Duct, and Pancreas Syndromes

Syndrome	Core mechanism	Clinical clue	Diagnostic anchor	Management anchor
Biliary colic	Transient cystic duct obstruction	Episodic postprandial RUQ/epigastric pain	RUQ ultrasound	Elective cholecystectomy if recurrent/symptomatic
Acute cholecystitis	Persistent cystic duct obstruction with inflammation	Persistent RUQ pain, fever, Murphy sign, leukocytosis	Ultrasound; HIDA if equivocal	Admission, IV fluids, antibiotics, early cholecystectomy or drainage if high risk

Syndrome	Core mechanism	Clinical clue	Diagnostic anchor	Management anchor
Choledocholithiasis	Stone in common bile duct	Jaundice, cholestatic labs, duct dilation	MRCP/EUS/ERCP by probability	ERCP with sphincterotomy/extraction if confirmed/high probability
Cholangitis	Infected obstructed bile duct	Charcot triad; Reynolds pentad if shock/confusion	Labs + ultrasound/CT/MRCP; ERCP for therapy	IV antibiotics, fluids, urgent biliary decompression
Acute pancreatitis	Pancreatic inflammation from gallstones, alcohol, hypertriglyceridemia, drugs, etc.	Epigastric pain radiating to back, nausea/vomiting	2 of 3: pain, lipase/amylase $\geq 3\times$ ULN, imaging	Early fluids, analgesia, nutrition, treat cause
Pancreatic cancer	Malignant ductal obstruction/invasion	Painless jaundice, weight loss, new diabetes, back pain	Pancreatic protocol CT; tissue diagnosis if needed	Resection if localized; palliation/systemic therapy if advanced

5. High-Yield Numeric Anchors

Numeric anchors do not replace clinical judgment, but they are powerful retrieval nodes. They help connect a symptom to a threshold, a diagnostic category, or a management branch.

Number	Clinical meaning
Hemoglobin about 7 g/dL	Common restrictive transfusion threshold for many hospitalized patients with upper GI bleeding after individualized assessment.
Glasgow-Blatchford 0-1	Very-low-risk upper GI bleeding category often considered for outpatient management after clinician assessment.
Age 45-75	Average-risk colorectal cancer screening age range in USPSTF guidance; ages 76-85 individualized.
SAAG ≥ 1.1 g/dL	Ascites likely due to portal hypertension physiology.
Ascitic PMN ≥ 250 cells/mm ³	Diagnostic threshold for spontaneous bacterial peritonitis treatment in the right clinical setting.
INR >1.5 plus encephalopathy	Core acute liver failure criterion when no preexisting cirrhosis is present.
Lipase/amylase $\geq 3\times$ upper limit	One of the diagnostic criteria for acute pancreatitis when paired with typical pain or imaging.
Common bile duct dilation	Raises probability of choledocholithiasis/obstruction when paired with jaundice or cholestatic labs.
Cecal diameter >10 -12 cm	Danger zone for perforation risk in severe colonic dilation/pseudo-obstruction/toxic megacolon context.
UC pancolitis duration ≥ 8 years	Common anchor for beginning colorectal cancer surveillance in extensive colitis; timing varies by guideline and risk factors.
B12 and bile acid absorption in terminal ileum	Explains Crohn-related B12 deficiency, bile acid diarrhea, gallstone risk, and post-ileal resection complications.
Hepatitis B serology triad	HBsAg means current infection; anti-HBs means immunity; anti-HBc indicates prior exposure or infection context.
Acetaminophen toxicity	Treat suspected significant overdose or acute liver injury with N-acetylcysteine promptly; do not wait for severe injury if risk is high.

6. Organ-Based Rapid Review

Esophagus and Stomach

Dysphagia is best split into difficulty with solids only versus solids and liquids. Solids-only dysphagia suggests a mechanical narrowing such as cancer, peptic stricture, Schatzki ring, eosinophilic esophagitis, or extrinsic compression. Progressive solids-to-liquids dysphagia with weight loss is cancer until proven otherwise. Intermittent dysphagia to solids with a ring-like lesion suggests Schatzki ring. Dysphagia to both solids and liquids from the beginning suggests a motility disorder such as achalasia or diffuse esophageal spasm. Achalasia produces impaired LES relaxation and aperistalsis; barium swallow may show a bird-beak pattern, but manometry is the key physiologic confirmation.

GERD becomes clinically important when symptoms are frequent, complicated, or accompanied by alarm features. Alarm features include dysphagia, odynophagia, weight loss, GI bleeding, iron deficiency anemia, persistent vomiting, or

suspected malignancy. Barrett esophagus is intestinal metaplasia from chronic reflux and increases adenocarcinoma risk. Peptic ulcer disease is usually driven by H. pylori, NSAID injury, or acid hypersecretion. Gastric ulcers require biopsy to exclude malignancy, whereas uncomplicated duodenal ulcers are less suspicious for cancer. Upper GI bleeding management begins with resuscitation, medication review, risk stratification, and endoscopy.

Small Bowel and Malabsorption

The small bowel is the main organ of nutrient absorption, so small bowel disease often presents as diarrhea, weight loss, bloating, anemia, vitamin deficiency, or electrolyte disturbance rather than localized pain alone. Celiac disease is an immune response to gluten-associated antigens in genetically susceptible patients and can cause iron deficiency, folate deficiency, osteoporosis from vitamin D deficiency, dermatitis herpetiformis, and chronic diarrhea. Testing is most reliable while the patient is still eating gluten. Small intestinal bacterial overgrowth causes bloating, diarrhea, malabsorption, and B12 deficiency due to abnormal bacterial metabolism in the small intestine.

Small bowel obstruction is a mechanical syndrome of proximal dilation, vomiting, dehydration, and electrolyte loss. Adhesions are a common cause in adults, with hernias, malignancy, Crohn strictures, volvulus, and intussusception as important alternatives. In contrast, ileus is decreased motility without a fixed mechanical blockage and is associated with surgery, opioids, anticholinergics, hypokalemia, severe illness, peritonitis, and spinal cord injury. Obstruction with fever, leukocytosis, acidosis, continuous pain, peritonitis, or shock suggests strangulation or ischemia and requires urgent surgical evaluation.

Colon, Rectum, and Vascular Emergencies

Colonic disease often presents with altered bowel habits, hematochezia, occult blood, anemia, LLQ pain, or constipation/obstruction. Colorectal cancer usually arises from adenomatous polyps through accumulation of molecular changes. Right-sided colon cancer often presents with occult bleeding and iron deficiency because the lumen is wider and stool is liquid. Left-sided tumors more often cause change in stool caliber, obstruction, and visible blood. Rectal cancer adds tenesmus, incomplete evacuation, and local pelvic symptoms.

Diverticulosis is common with age and is usually asymptomatic, but it can cause painless lower GI bleeding or diverticulitis. Diverticulitis is inflammation/infection around a diverticulum and presents with LLQ pain, fever, leukocytosis, and sometimes abscess, perforation, fistula, or obstruction. Angiodysplasia is a vascular ectasia classically associated with older adults and can cause recurrent painless bleeding. Acute mesenteric ischemia is an emergency defined by intestinal hypoperfusion; pain out of proportion to exam, atrial fibrillation, atherosclerosis, low-flow shock, or hypercoagulability should trigger urgent evaluation.

Inflammatory and Functional Bowel Disease

IBD is inflammatory and structural; IBS is functional and does not cause mucosal ulceration, anemia, fever, hypoalbuminemia, nocturnal symptoms, or objective inflammation. Crohn disease is transmural, patchy, and can involve any segment from mouth to anus, which explains fistulas, abscesses, strictures, perianal disease, and obstruction. Ulcerative colitis is continuous mucosal disease beginning in the rectum, which explains bloody diarrhea, urgency, tenesmus, toxic megacolon risk, and a colon cancer risk that rises with duration and extent of colitis.

Extraintestinal manifestations connect GI inflammation to the eyes, skin, joints, biliary tract, and clotting system. Episcleritis and some peripheral arthritis can parallel bowel activity, while ankylosing spondylitis and primary sclerosing cholangitis may follow independent courses. Fecal calprotectin is useful because neutrophil-derived intestinal inflammation makes IBD more likely and IBS less likely. Colonoscopy with biopsy remains central when IBD, microscopic colitis, malignancy, or unexplained inflammatory diarrhea is suspected.

Liver and Portal Hypertension

Liver disease must be described by injury pattern, duration, severity, and complications. Acute liver failure is not just high aminotransferases; it is severe acute hepatic injury with encephalopathy and impaired synthetic function, classically INR >1.5, in a patient without established cirrhosis. Cirrhosis is chronic architectural distortion with fibrosis and regenerative nodules. Portal hypertension produces ascites, varices, splenomegaly, thrombocytopenia, caput medusae, and portosystemic shunting. Synthetic failure produces coagulopathy, hypoalbuminemia, and impaired detoxification.

Ascites should be tapped when new, clinically changed, infected, or associated with hospitalization or decompensation. SAAG links ascites to portal pressure physiology, while ascitic PMN count determines SBP treatment. Hepatic encephalopathy is a clinical syndrome of impaired cognition, sleep-wake reversal, asterixis, and sometimes coma; common triggers include GI bleeding, infection, constipation, renal failure, dehydration, electrolyte disturbance, sedatives, and excess dietary protein in susceptible patients. Treatment connects lactulose, rifaximin, trigger correction, and avoidance of oversedation.

Biliary Tract and Pancreas

Biliary disease is organized by the location and consequences of obstruction. Cystic duct obstruction causes biliary colic when transient and acute cholecystitis when persistent with gallbladder inflammation. Common bile duct obstruction causes choledocholithiasis and obstructive jaundice. Infection behind obstruction causes ascending cholangitis and can progress rapidly to septic shock. PSC produces multifocal biliary strictures and is linked to ulcerative colitis and cholangiocarcinoma. PBC is autoimmune destruction of small intrahepatic bile ducts, classically in middle-aged women, with cholestatic labs, pruritus, fatigue, and antimitochondrial antibodies.

Pancreatitis requires two of three diagnostic features: characteristic pain, pancreatic enzymes at least three times the upper limit of normal, or imaging evidence. Gallstones and alcohol are common causes; hypertriglyceridemia, hypercalcemia, medications, ERCP, trauma, and malignancy are also important. Chronic pancreatitis causes recurrent pain, calcifications, exocrine insufficiency, steatorrhea, diabetes, pseudocysts, and pancreatic cancer risk. Pancreatic cancer should be suspected with painless jaundice, weight loss, new-onset diabetes in an older adult, migratory thrombophlebitis, or a pancreatic mass.

7. Mixed Integrated Cases

Case 1: Melena after NSAID Use

Case stem. A 68-year-old man with osteoarthritis has several days of black tarry stools and lightheadedness. He takes naproxen daily and aspirin after a prior MI. He is tachycardic but alert. Hemoglobin is 7.2 g/dL.

Question. What is the most likely source and initial management strategy?

Answer. Upper GI bleeding from peptic ulcer disease or erosive disease is likely. Stabilize with IV access, type/cross, labs, PPI therapy, medication review, and upper endoscopy after stabilization.

Teaching point. Melena usually indicates blood exposed to intestinal bacteria for hours and most often reflects an upper GI source. NSAIDs and aspirin impair mucosal defenses and platelet function, increasing ulcer bleeding risk.

Case 2: RUQ Pain with Sepsis

Case stem. A 74-year-old woman has fever, jaundice, RUQ pain, confusion, and hypotension. Labs show direct hyperbilirubinemia and elevated alkaline phosphatase.

Question. What diagnosis requires urgent source control?

Answer. Ascending cholangitis with sepsis. Treat with IV fluids, blood cultures, broad-spectrum antibiotics, and urgent biliary decompression, usually by ERCP once stabilized.

Teaching point. Charcot triad is fever, jaundice, and RUQ pain. Reynolds pentad adds hypotension and altered mental status, signaling severe disease and high mortality risk without drainage.

Case 3: Chronic Diarrhea with Red Flags

Case stem. A 29-year-old woman has 4 months of diarrhea, nocturnal stools, weight loss, oral ulcers, and RLQ pain. Hemoglobin is low, CRP is elevated, and fecal calprotectin is high.

Question. IBS or IBD?

Answer. IBD is much more likely, particularly Crohn disease involving the ileocecal region. Colonoscopy with ileal intubation and biopsy plus cross-sectional enterography may be needed.

Teaching point. Nocturnal diarrhea, anemia, weight loss, objective inflammation, oral ulcers, and focal RLQ pain are not typical IBS features. Terminal ileal Crohn links to B12 deficiency, bile acid diarrhea, strictures, and fistulas.

Case 4: New Ascites in Cirrhosis

Case stem. A patient with known cirrhosis is admitted with increasing abdominal distention and mild confusion. Temperature is 37.9 C. Creatinine is increased from baseline.

Question. What bedside diagnostic procedure should not be delayed?

Answer. Diagnostic paracentesis. Send cell count with differential, albumin/protein, culture, and other tests guided by the presentation.

Teaching point. SBP can present subtly. An ascitic PMN count ≥ 250 cells/mm³ is enough to treat presumptively in the appropriate setting. Renal injury and mental status change increase concern.

Case 5: Pain Out of Proportion

Case stem. A 79-year-old man with atrial fibrillation develops abrupt severe abdominal pain. He is uncomfortable, but the abdomen is initially soft with minimal tenderness. Lactate is elevated.

Question. What is the dangerous diagnosis?

Answer. Acute mesenteric ischemia, likely embolic SMA occlusion. Obtain urgent vascular/surgical consultation and CT angiography if stable enough; begin resuscitation, antibiotics, and anticoagulation when appropriate.

Teaching point. Early mesenteric ischemia can have severe visceral pain before peritoneal signs develop. Peritonitis, acidosis, shock, or free air suggests infarction or perforation.

Case 6: Painless Jaundice

Case stem. A 66-year-old man has progressive painless jaundice, pale stools, dark urine, pruritus, weight loss, and a distended nontender gallbladder on exam.

Question. Which malignant process is concerning?

Answer. Pancreatic head cancer or distal cholangiocarcinoma causing extrahepatic biliary obstruction. Pancreatic-protocol CT and specialist evaluation are appropriate.

Teaching point. Painless obstructive jaundice with weight loss is malignant until proven otherwise. Gallstones usually cause painful episodic obstruction, while cancer produces progressive obstruction.

Case 7: Dysphagia Pattern

Case stem. A 58-year-old man has progressive dysphagia that began with meat and bread and now includes liquids. He has lost 8 kg and has a long tobacco history.

Question. What should be prioritized?

Answer. Esophageal cancer evaluation with endoscopy and biopsy. Barium swallow may help characterize dysphagia, but tissue diagnosis is essential.

Teaching point. Progressive solids-to-liquids dysphagia with weight loss is a mechanical narrowing until proven otherwise. Motility disorders usually cause solids-and-liquids dysphagia from the start.

Case 8: UC Flare or Infection

Case stem. A patient with known ulcerative colitis has worsening bloody diarrhea and fever after a recent antibiotic course.

Question. What infection must be excluded before escalating immunosuppression?

Answer. Clostridioides difficile infection. Test stool and evaluate severity; severe colitis requires careful monitoring for toxic megacolon.

Teaching point. IBD flares and infectious colitis overlap. Escalating steroids or biologics before excluding *C. difficile* can worsen outcomes.

8. Practice Questions

Question 1. A 43-year-old woman has chronic watery diarrhea and a normal colonoscopy appearance. Random biopsies show lymphocytic inflammation. Diagnosis?

Answer. Microscopic colitis. Gross colonoscopy can be normal; diagnosis requires biopsy.

Question 2. A cirrhotic patient vomits blood. What two empiric medication categories are often started before endoscopy when variceal bleed is suspected?

Answer. A vasoactive agent such as octreotide and antibiotics, alongside resuscitation and urgent endoscopic therapy.

Question 3. A patient has epigastric pain radiating to the back and lipase 5 times upper limit of normal. What are the first management priorities?

Answer. Fluids, analgesia, early enteral nutrition as tolerated, severity assessment, and evaluation for cause such as gallstones, alcohol, triglycerides, calcium, and medications.

Question 4. Which IBD complication is classically associated with ulcerative colitis and can occur independent of bowel activity?

Answer. Primary sclerosing cholangitis.

Question 5. A patient with chronic liver disease has ascitic fluid albumin 0.8 g/dL and serum albumin 2.4 g/dL. What does the SAAG suggest?

Answer. SAAG is 1.6 g/dL, supporting portal hypertension physiology.

Question 6. Which colorectal cancer presentation is more typical of right-sided tumors?

Answer. Occult bleeding, iron deficiency anemia, fatigue, and later obstruction due to larger lumen.

Question 7. What is the difference between ileus and small bowel obstruction?

Answer. Ileus is impaired motility without a fixed mechanical blockage and often shows gas throughout bowel; SBO is mechanical obstruction with proximal dilation and distal decompression.

Question 8. What is the first-line imaging test for suspected acute cholecystitis?

Answer. Right upper quadrant ultrasound; HIDA can help when ultrasound is equivocal.

Question 9. A patient has HBsAg negative, anti-HBs positive, anti-HBc negative. Interpretation?

Answer. Immunity from vaccination.

Question 10. A young patient has liver disease, movement disorder, psychiatric symptoms, and low ceruloplasmin. Diagnosis?

Answer. Wilson disease; confirm with appropriate copper studies and ophthalmologic evaluation for Kayser-Fleischer rings.

9. Graph-RAG Concept Extraction Map

The table below is written to support concept extraction. A retrieval system can convert each row into nodes and edges: symptom-to-diagnosis, diagnosis-to-test, diagnosis-to-treatment, disease-to-complication, and risk-factor-to-disease relationships.

Central concept	High-value connected nodes	Relationship examples
Melena	Upper GI bleeding, PUD, varices, small bowel bleeding, BUN elevation, endoscopy, anemia	Melena suggests upper GI bleeding; PUD causes melena; EGD evaluates melena.

Central concept	High-value connected nodes	Relationship examples
Hematochezia	Diverticular bleed, angiodysplasia, colorectal cancer, ischemic colitis, brisk UGIB, colonoscopy, CTA	Hematochezia usually localizes to lower GI but can reflect brisk upper GI bleeding.
Dysphagia	Esophageal cancer, achalasia, stricture, Schatzki ring, EoE, manometry, endoscopy	Progressive solids-to-liquids dysphagia raises cancer concern; solids-and-liquids from onset raises motility concern.
Chronic diarrhea	IBS, IBD, celiac disease, microscopic colitis, SIBO, pancreatic insufficiency, bile acid diarrhea	Nocturnal diarrhea and anemia argue against IBS; celiac disease causes malabsorption.
Jaundice	Hemolysis, hepatitis, cirrhosis, choledocholithiasis, cholangitis, pancreatic cancer, cholangiocarcinoma	Direct hyperbilirubinemia and alkaline phosphatase suggest cholestasis; ultrasound evaluates obstruction.
Ascites	Portal hypertension, cirrhosis, SAAG, SBP, malignancy, heart failure, diuretics, paracentesis	SAAG ≥ 1.1 links ascites to portal hypertension; PMN ≥ 250 indicates SBP treatment.
RUQ pain	Biliary colic, cholecystitis, cholangitis, hepatitis, hepatic abscess, pneumonia, PE	Fever plus RUQ pain plus jaundice suggests cholangitis; ultrasound is first-line for gallbladder disease.
Epigastric pain	PUD, pancreatitis, MI, biliary disease, gastric cancer, functional dyspepsia	Pain radiating to back and lipase elevation supports pancreatitis; NSAIDs raise PUD risk.
Cirrhosis	Portal hypertension, varices, ascites, encephalopathy, HRS, SBP, HCC, MELD, Child-Pugh	Cirrhosis causes portal hypertension; cirrhosis increases HCC risk; MELD estimates transplant priority.
IBD	Crohn, UC, calprotectin, colonoscopy, steroids, biologics, toxic megacolon, CRC surveillance	Crohn causes fistulas; UC causes continuous rectal colitis; IBD increases thromboembolism risk.

10. Selected Guideline Anchors and Source Coverage

This chapter is an original Medvale synthesis. The uploaded GI source PDFs and previously generated GI chapters were used as topic coverage signals only. The following clinical anchors were checked against current or major guideline sources: colorectal cancer screening starting at age 45 for average-risk adults, restrictive transfusion thresholds around hemoglobin 7 g/dL in many hospitalized upper GI bleeding patients, and AASLD guidance for ascites/SBP/HRS. Clinical decisions should always be individualized to patient context, institutional protocols, and attending/specialist recommendations.

Anchor	Reference note
Colorectal cancer screening	USPSTF recommends screening average-risk adults ages 45-75, with ages 76-85 individualized.
Upper GI bleeding	ACG guideline suggests restrictive RBC transfusion threshold around 7 g/dL for hospitalized UGIB patients and endoscopy within 24 hours after presentation/stabilization.
Ascites/SBP/HRS	AASLD practice guidance provides diagnostic and management frameworks for ascites, spontaneous bacterial peritonitis, and hepatorenal syndrome.
Prior Medvale GI coverage	Chapters 3A-3H: overview, esophagus/stomach, small bowel/malabsorption, colon/LGIB/mesenteric disease, IBD/IBS/colitis, liver/cirrhosis/masses, biliary/pancreas/appendix, viral/metabolic/alcohol/DILI liver disease.